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Chandigarh, India 160101

WEBSITES, PORTFOLIO PROFILES

- www.linkedin.com/in/rosh-an-singh-26a678117
- <https://rao-extractor-surf-analysis.onrender.com>
- <https://github.com/SScofieldd>

EDUCATION

M.Tech, Ocean Technology
IIT Madras | 8.96 CGPA, Chennai
January 2021
Prof K A V Pandalai Prize (Silver Medal) for academic performance.

B.Tech, Mechanical Engineering
KIET Institute of Technology | 75.32%, Ghaziabad
January 2018
Graduated with Honors

12TH, Science
Kendriya Vidyalaya No.1 | 85%, Army, Jodhpur
January 2013
Certificate of Merit for outstanding academic performance, ranking in the top 0.1% in Physics (All India SR. SCH Certification 2013)

ROSHAN SINGH

Sr. Engineering Analyst

PROFESSIONAL SUMMARY

Senior Engineering Analyst with 4.5 years of experience in global and local analysis of dynamic subsea umbilical systems using OrcaFlex and UFlex. Experienced in static and dynamic analysis, engineering response assessment, technical documentation, task planning and analysis delivery coordination for offshore projects. Currently coordinating analysis delivery for the Bacalhau project and collaborating with international engineering teams. Proficient in OrcaFlex, OrcFxAPI, Python and MATLAB, with directly transferable experience relevant to dynamic riser engineering.

TECHNICAL SKILLS

- **Engineering Analysis:** Dynamic Umbilicals, Global Analysis, Local Analysis, Static and Dynamic Analysis, Engineering Response Assessment
- **Software:** OrcaFlex, UFlex, OrcFxAPI, Python, MATLAB, PyQt, Microsoft Excel
- **Project Delivery:** Technical Documentation, Load-Case Management, Task Planning, Progress Tracking, Multidisciplinary Coordination
- **Automation:** Engineering Workflow Automation, Data Processing, Result Extraction and Reporting

WORK EXPERIENCE

2022 - Present

Oceaneering - Sr. Engineering Analyst, Chandigarh

- Perform global and local analysis of dynamic subsea umbilical systems using OrcaFlex and UFlex for offshore engineering applications.
- Develop and review static and dynamic analysis models, load cases, engineering responses and technical deliverables.
- Coordinate Bacalhau project analysis delivery, including task planning, resource allocation, progress tracking, technical issue resolution and timely issuance of deliverables.
- Automated analyses, saving ~2,600 hours and enhancing team efficiency.
- Implemented ML-driven crush prediction, reducing analysis time from two weeks to 20 seconds.
- Developed & enhanced the MATLAB-based OptiMum tool, adopted globally.

- Received the Excellence Award for automation, innovation, and efficiency.
 - Collaborated with international engineering teams and visited Oceaneering's Rosyth facility in Scotland for knowledge transfer, engineering-tool development, and alignment of analysis practices.
 - Develop engineering tools and automated workflows using Python, MATLAB, OrcFxAPI, PyQt and Excel for OrcaFlex model generation, simulation execution, result extraction, data validation, design calculations and reporting.
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ACCOMPLISHMENTS

- Certificate of Excellence for successful completion of the Trident Energy Okume - Ceiba Subsea Tie-Back Project, recognising technical contribution and project delivery.
 - Productivity Excellence Award for engineering automation initiatives that delivered an estimated saving of approximately 2,600 engineering hours.
 - Team Excellence Award
 - Prof. K. A. V. Pandalai Prize (Silver Medal), IIT Madras, for academic performance.
 - Certificate of Merit for outstanding academic performance, ranking in the top 0.1% in Physics (All India SR. SCH Certification 2013)
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INTERESTS

Painting | Fitness Training | Music | Traveling | Movies, Continuous learning exploring new skills like cooking, video editing, and evolving interests over time.

ABOUT ME

Beyond my core engineering responsibilities, I have taken ownership of complex subsea analysis activities, including the Bacalhau project, where I support analysis planning, technical documentation, task coordination and delivery. I also contributed to the successful completion of the Trident Energy Okume-Ceiba Subsea Tie-Back project, for which I received a Certificate of Excellence. I have a strong interest in engineering automation and have developed OrcaFlex, UFlex, Python and MATLAB-based tools for simulation, design calculations, post-processing and reporting. My UFlex automation initiative alone delivered approximately 2,600 hours of productivity savings and earned me a Productivity Excellence Award.

Outside work, I enjoy painting, fitness training, music, travelling and continuous learning. These interests help me stay creative, disciplined and open to new ideas.